Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Assignment

Subject : P21A1MSD : MERN STACK DEVELOPMENT

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: Prof. Vidhi H. Sheth | Email ID: vidhi.sheth@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)	Practical List	Mini Project / Case Studies
Unit-1: ReactJS Fundamentals:	1	Define ReactJS and explain its key features.	CO1	1,2,3,4,5,6,7,8,11	BTL-1	Build a React application using Functional Components, JSX and Component Reusability concepts.	Mini Project: Design and develop a full-stack industry-oriented MERN application (Document Management System).
	2	Explain JSX and its advantages in ReactJS.	CO1	1,2,3,4,5,6,7,8,11	BTL-2	Develop a React application implementing Props, State Management and Event Handling.	
	3	Differentiate between ReactJS, Angular, and Vue.	CO1	1,2,3,4,5,6,7,8,11	BTL-4	Create a Student Management System using React Forms, Controlled Components and Validation.	
	4	Explain the concept of Virtual DOM with suitable examples.	CO1	1,2,3,4,5,6,7,8,11	BTL-2	Develop a Single Page Application (SPA) using React Router for navigation between multiple pages.	
	5	Create a React application demonstrating Class and Functional Components.	CO1	1,2,3,4,5,6,7,8,11	BTL-3	Develop a React application consuming REST APIs using Axios and Fetch API.	
Unit-2: Forms, Events and Routing in ReactJS	7	Explain the lifecycle phases of a React component.	CO2	1,2,3,4,5,6,7,8,11	BTL-2	Create a backend server using Express.js implementing routing, middleware and request handling.	
	8	Differentiate between Props and State in ReactJS.	CO2	1,2,3,4,5,6,7,8,11	BTL-4	Build RESTful APIs for Student Records Management using Express.js and Postman testing.	
	9	Explain controlled and uncontrolled components with examples.	CO2	1,2,3,4,5,6,7,8,11	BTL-2	Develop an Employee Management API integrating Express.js with MongoDB Atlas.	
	10	Demonstrate event handling in ReactJS.	CO2	1,2,3,4,5,6,7,8,11	BTL-3	Create a file upload module using Multer and store file metadata in MongoDB.	
	11	Develop a React application using forms and routing.	CO2	1,2,3,4,5,6,7,8,11	BTL-6	Develop a complete MERN application integrating React Frontend, Express Backend and MongoDB Database.	
Unit-3: Node.js	13	Explain the architecture and features of Node.js.	CO3	1,2,3,4,5,6,7,8,11	BTL-2	Create a Blog Management System implementing CRUD Operations, Authentication and Role-Based Access Control.	
	14	Describe the working of REPL in Node.js.	CO3	1,2,3,4,5,6,7,8,11	BTL-2	Develop a Real-Time Chat Application using Socket.IO, React and Node.js.	
	15	Explain Node Package Manager (NPM) and its advantages.	CO3	1,2,3,4,5,6,7,8,11	BTL-2		
	16	Demonstrate File System operations in Node.js.	CO3	1,2,3,4,5,6,7,8,11	BTL-3		
	17	Develop a Node.js application using HTTP module and EventEmitter.	CO3	1,2,3,4,5,6,7,8,11	BTL-6		
Unit 04: Cascading Style Sheets	19	Explain the features and architecture of ExpressJS.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	20	Describe routing and middleware in ExpressJS.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	21	Differentiate between SQL and NoSQL databases.	CO4	1,2,3,4,5,6,7,8,11	BTL-4		
	22	Explain various types of NoSQL databases with examples.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	23	Perform CRUD operations using MongoDB.	CO4	1,2,3,4,5,6,7,8,11	BTL-3		
Unit 05: JavaScript	25	Explain API and REST API architecture.	CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	26	Describe the process of connecting Node.js with MongoDB.	CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	27	Explain ReactJS and Node.js integration.	CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	28	Implement a REST API using Node.js and ExpressJS.	CO5	1,2,3,4,5,6,7,8,11	BTL-3		
	29	Design a complete MERN Stack application integrating React, Node.js, ExpressJS, and MongoDB.	CO5	1,2,3,4,5,6,7,8,11	BTL-6		

Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Assignment

Subject : PYAA2WDT : WEB DEVELOPMENT TECHNOLOGIES

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: SHUBHAM D MEVADA | Email ID: shubham.mevada@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)	Practical List	Mini Project / Case Studies
Unit-1: Basics of PHP	1	Define PHP. Explain its role in server-side web application development.	CO1			1.Develop a PHP program to demonstrate basic syntax, variables, constants, data types, operators, input/output, and debugging using echo, print, and var_dump().	Create a Backend System for an Event Booking System with QR Code and Online Payment Gateway Integration (Razorpay Demo Mode Only)
	2	Explain the client-server working model of PHP with a neat diagram.	CO1			2.Develop a Student Grade Calculator using decision-making statements (if, if-else, switch) and looping constructs (for, while, foreach) to generate reports and patterns.	
	3	Differentiate between client-side scripting and server-side scripting with suitable examples.	CO1			3.Develop reusable PHP functions for student result processing, including marks calculation, grade generation, string validation, and array manipulation.	
	4	Explain PHP syntax, variables, constants, and data types with examples.	CO1			4.Design a multi-page Student Registration module using HTML forms with GET and POST methods, including server-side validation.	
	5	Discuss various control structures and user-defined functions available in PHP.	CO1			5.Develop a Student Record Management module using indexed arrays, associative arrays, multidimensional arrays, and built-in array functions.	
	6	Explain arrays, string manipulation functions, and date/time functions in PHP with suitable examples.	CO1			6.Develop a Student Information System using Object-Oriented PHP by implementing classes, objects, constructors, methods, encapsulation, and static members.	
Unit-2: Working with OPP and Advance OOP	1	Explain the basic concepts of Object-Oriented Programming in PHP with suitable examples.	CO2			7.Extend the Student Information System by implementing inheritance, abstract classes, interfaces, traits, and method overriding to model different user roles (Student, Faculty, Administrator).	
	2	Discuss constructors, destructors, inheritance, and polymorphism in PHP.	CO2			8.Develop a secure Login and Logout system using Sessions, Cookies, URL Rewriting, and Hidden Form Fields for user authentication and state management.	
	3	Explain abstract classes, interfaces, and traits in PHP with examples.	CO2			9.Create a File Management module to upload, download, preview, and delete student profile images and documents with appropriate validation.	
	4	Describe the process of file uploading and downloading in PHP applications.	CO2			10.Establish a MySQL database connection using PDO and create database tables for the Student Information System.	
	5	Explain session management in PHP and compare Sessions with Cookies.	CO2			11.Develop CRUD operations (Add, View, Edit, Delete) for student records using PDO Prepared Statements to ensure secure database access.	
	6	Discuss various state management techniques such as URL Rewriting and Hidden Form Fields with examples.	CO2			12.Enhance the Student Information System by implementing searching, sorting, filtering, pagination, and record updates using PDO.	
Unit-3: Working with data using PDO, XML and JSON	1	Define PDO. Explain its advantages over traditional database connectivity methods in PHP.	CO3			13.Develop a CSV Import and Export module to manage student records between MySQL and spreadsheet applications.	
	2	Describe the steps involved in installing and configuring PDO drivers.	CO3			14.Develop XML-based data storage by exporting student records to XML and reading XML data back into the application.	
	3	Explain CRUD operations using PDO with suitable examples.	CO3			15.Develop REST-style JSON services to expose student records and create a PHP client application to consume JSON data.	
	4	Discuss XML and XML Parser in PHP. Explain how XML data can be read and written.	CO3			16.Integrate SMTP email functionality to send registration confirmations, password reset links, and admission notifications	
	5	Explain the concept of Web Services and their role in web application integration.	CO3			17.Integrate third-party APIs to send SMS notifications and implement social login (Google/Facebook) for user authentication.	
	6	Define JSON. Explain JSON encoding and decoding functions in PHP with examples.	CO3			18.Enhance the Student Information System by implementing secure coding practices including server-side validation, password hashing, SQL Injection prevention, Cross-Site Scripting (XSS) protection, and file upload security.	
Unit 04: Payment Gateway, Email, Social Media	1	Explain the architecture and working of a Payment Gateway in web applications.	CO4			19.Deploy the Student Information System on a web server using GitHub for version control, FTP/FileZilla for deployment, and implement database backup and restoration.	
	2	Discuss the steps involved in integrating an online payment gateway using PHP.	CO4			20.Develop a complete Student Information System by integrating authentication, profile management, file uploads, database operations, JSON services, email notifications, and security features into a single web application.	
	3	Explain how email services can be integrated into PHP web applications using SMTP.	CO4				
	4	Discuss the integration of SMS services in PHP applications using third-party APIs.	CO4				
	5	Explain social media authentication and its advantages in modern web applications.	CO4				
	6	Discuss the process of integrating social media networks into a PHP web application.	CO4				
Unit 05: Web Security, Deployment and Backup of Web Application	1	Explain the importance of data validation and sanitization in web applications.	CO5				
	2	Define SQL Injection. Discuss its impact and prevention techniques using PDO Prepared Statements.	CO5				
	3	What is Cross-Site Scripting (XSS)? Explain different types of XSS attacks and their prevention.	CO5				
	4	Describe the process of deploying a PHP web application on a production server.	CO5				
	5	Explain the procedure for creating database backup and restore scripts in PHP.	CO5				
	6	Discuss the role of FTP, GitHub, and FileZilla in web application management and deployment.	CO5				

Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Assignment

Subject : P21A1MSD : MERN STACK DEVELOPMENT

BTL1 -Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: Prof. Vidhi H. Sheth | Email ID: vidhi.sheth@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)	Practical List	Mini Project / Case Studies
Unit: 01 - Digital Marketing:	1	Elaborate the need of digital marketing in current scenario. OR Describe the importance of Digital Marketing for Business.	CO1	1,2,3,4,5,6,7,8,11	BTL-1	**01. Write steps to create Local Listing on Google My Business with Fig.** #What is Local Listing #List of Local Listing Sites #Important of Local Listing Sites #Features of Local Listing Sites.	
	2	Explain the Different ways/Types for digital marketing in details.	CO1	1,2,3,4,5,6,7,8,11	BTL-2	**02. Create a Landing Page for Upcoming Workshop with Registration Form Using Google Sites/canva / AI Platforms. (Workshop : AI Tools Date: 21 Aug 2026, Expert : Prof. Ravi Patel)** #What is Landing Page #List of Landing Page Development Platforms #Features of Landing Page #Importance of Landing Page.	
	3	What is Digital Marketing ? Listout the Objective of digital marketing.	CO1	1,2,3,4,5,6,7,8,11	BTL-4	03. Create a facebook Business page, Instgram account, Thread account, Business Whatsapp & Integrate it for meta campaign.	
	4	Describe the importance of Landing Page with example. Explain the Features of different Platforms where we can create a landing pages.	CO1	1,2,3,4,5,6,7,8,11	BTL-2	04. Execute the **Lead Generation** Meta Campaign for AI Tools Workshop.	
	5	What is Local Listing ? Explain the importance, requirement, and features of different Local Listing Sites.	CO1	1,2,3,4,5,6,7,8,11	BTL-3	05. Execute the meta Traffic Campaign for Workshop Registration.	
	7	Discuss the features of Google My business Local Listing Sites in details with example.	CO2	1,2,3,4,5,6,7,8,11	BTL-2	06. Write steps to create different types of Meta Ads Forms for Lead Generation Campaign.	
	8	State the difference between Website & Landing Page.	CO2	1,2,3,4,5,6,7,8,11	BTL-4		
	9	Elaborate the types of Meta Campaigns with example.	CO2	1,2,3,4,5,6,7,8,11	BTL-2		
	10	Explain the basic technical requirement of meta Ads execution.	CO2	1,2,3,4,5,6,7,8,11	BTL-3		
	11	State the difference between Organic & Paid Digital Marketing.	CO2	1,2,3,4,5,6,7,8,11	BTL-6		
	13	State the importance of Social Media platforms in current scenario.	CO3	1,2,3,4,5,6,7,8,11	BTL-2		
	14	Elaborate the Lead generation Campaigns with example.	CO3	1,2,3,4,5,6,7,8,11	BTL-2		
	15	Write steps to create lead Generation Campaign for Workshop Registration.	CO3	1,2,3,4,5,6,7,8,11	BTL-2		
	16	Discuss the importance of Traffic Campaign with its steps & Example.	CO3	1,2,3,4,5,6,7,8,11	BTL-3		
	17	Describe the Role of LinkEdin for PG students.	CO3	1,2,3,4,5,6,7,8,11	BTL-6		
	19	**Define the Following Terms of Digital Marketing:** ROI, Landing Page, Digital Marketing, SEO, Performance Marketing, Email Marketing, CTA (Call to Action), SERP, Keyword Research, On-Page SEO, Off-Page SEO, Technical SEO, Backlink, Internal Link, DA, PA, Indexing, Crawling, Ranking, PPC, Web Analytics, Bounce Rate, CRM.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	20	What is Email Marketing ? Discuss the Role of Email Marketing.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	21	Explain the Features of Email Marketing tools.	CO4	1,2,3,4,5,6,7,8,11	BTL-4		
	22	Describe the any three paid Email Marketing Platforms with its Features.	CO4	1,2,3,4,5,6,7,8,11	BTL-2		
	23	How Email Marketing is effective in current scenario.	CO4	1,2,3,4,5,6,7,8,11	BTL-3		
	25		CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	26		CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	27		CO5	1,2,3,4,5,6,7,8,11	BTL-2		
	28		CO5	1,2,3,4,5,6,7,8,11	BTL-3		
	29		CO5	1,2,3,4,5,6,7,8,11	BTL-6		

Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Assignment

Subject : P21A3PYT : PYTHON PROGRAMMING

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: Dr. Ajay Patel | Email ID: ajaykumar.patel@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions / Task Description	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)
1	1	Explain Python's memory management mechanism, including object mutability, dynamic typing, garbage collection via reference counting, and cycle detection.	CO1	1,2,3,4,5,6,7,8,11	BTL-2
1	2	Differentiate between dynamic typing in Python and static typing in C/Java. Analyze the impact of Python's Global Interpreter Lock (GIL) on multithreaded processing.	CO1	1,2,3,4,5,6,7,8,11	BTL-4
1	3	Compare memory layout and computational efficiency of native Python lists versus vectorized NumPy N-dimensional arrays with memory benchmarking examples.	CO1	1,2,3,4,5,6,7,8,11	BTL-4
1	4	Describe broadcasting rules in NumPy with explicit mathematical expressions and shape compatibility conditions across multi-dimensional arrays.	CO1	1,2,3,4,5,6,7,8,11	BTL-2
1	5	Differentiate between Pandas Series and DataFrame objects. How do loc, iloc, and indexing mechanics differ when slicing multi-index DataFrames?	CO1	1,2,3,4,5,6,7,8,11	BTL-2
2	6	Differentiate between import module, from module import *, and dynamic loading via importlib. Explain namespace pollution risks and module isolation.	CO2	1,2,3,4,5,6,7,8,11	BTL-2
2	7	Compare Python's sequence data structures (List, Tuple, Dictionary, Set) in terms of asymptotic time complexity (Big-O) for access, insertion, search, and deletion operations.	CO2	1,2,3,4,5,6,7,8,11	BTL-4
2	8	Explain Method Resolution Order (MRO) in Python using C3 Linearization algorithm under complex multiple inheritance scenarios.	CO2	1,2,3,4,5,6,7,8,11	BTL-2
2	9	Explain the lifecycle of custom Exception handling in Python. Compare built-in error handling with user-defined Exception chains (raise ... from ...).	CO2	1,2,3,4,5,6,7,8,11	BTL-2
2	10	Contrast Class variables vs Dynamic Instance variables, dynamic attribute modification via __getattr__ and __setattr__, and property decorators (@property).	CO2	1,2,3,4,5,6,7,8,11	BTL-4
3	11	Explain string immutability in Python and evaluate string concatenation techniques (+, .join(), StringIO) for dynamic large-scale text aggregation.	CO3	1,2,3,4,5,6,7,8,11	BTL-2
3	12	Differentiate between context managers (with statement) and explicit file stream management (open/close). How do __enter__ and __exit__ function internally?	CO3	1,2,3,4,5,6,7,8,11	BTL-2
3	13	Compare file dynamic access techniques: line-by-line streaming vs. memory-mapped file access (mmap) for multi-gigabyte log analysis.	CO3	1,2,3,4,5,6,7,8,11	BTL-4
3	14	Explain regular expression syntax in Python (re module) for greedy vs. non-greedy matching, lookahead/lookbehind assertions, and group substitution.	CO3	1,2,3,4,5,6,7,8,11	BTL-2
3	15	Contrast standard text file operations with structured file formats (CSV, TSV, JSON). How are delimiter escaping and quote handling resolved?	CO3	1,2,3,4,5,6,7,8,11	BTL-2
4	16	Explain the Event-Driven Programming Paradigm in GUI design. How does mainloop() process event queues in dynamic user applications?	CO4	1,2,3,4,5,6,7,8,11	BTL-2
4	17	Compare tkinter layout managers (pack(), grid(), place()) detailing dynamic resize handling, anchor positioning, and widget overlapping rules.	CO4	1,2,3,4,5,6,7,8,11	BTL-4
4	18	Explain event binding mechanisms in tkinter using .bind(). Distinguish between event types, event modifiers, and callback handler parameter passes.	CO4	1,2,3,4,5,6,7,8,11	BTL-2
4	19	Describe how nested frames (Frame, LabelFrame) and custom styled themes enhance GUI modularity and component separation.	CO4	1,2,3,4,5,6,7,8,11	BTL-2
4	20	Analyze techniques for asynchronous background task execution within tkinter apps to keep main UI threads responsive during long operations.	CO4	1,2,3,4,5,6,7,8,11	BTL-4
5	21	Define Artificial Intelligence and distinguish between Artificial Narrow Intelligence (ANI), General Intelligence (AGI), and Superintelligence (ASI).	CO5	1,2,3,4,5,6,7,8,11	BTL-2
5	22	Compare Supervised, Unsupervised, and Reinforcement Learning paradigms. Outline core business trade-offs for each paradigm.	CO5	1,2,3,4,5,6,7,8,11	BTL-4
5	23	Differentiate Regression tasks from Classification tasks in Machine Learning. Detail output continuous variables vs discrete categorical target variables.	CO5	1,2,3,4,5,6,7,8,11	BTL-2
5	24	Explain the Bias-Variance Tradeoff in predictive models. Analyze how Underfitting and Overfitting manifest during training and validation loss curves.	CO5	1,2,3,4,5,6,7,8,11	BTL-4
5	25	Detail evaluation metrics for classification (Accuracy, Precision, Recall, F1-Score, ROC-AUC) vs regression (MSE, RMSE, MAE, R-squared).	CO5	1,2,3,4,5,6,7,8,11	BTL-2

Department of Computer Science (FCA) - Ganpat University					
M.Sc.(CA & IT) - Semester I : Assignment					
Subject : P21A3PYT : PYTHON PROGRAMMING					
BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;					
Subject Teacher: Dr. Ajay Patel | Email ID: ajaykumar.patel@ganpatuniversity.ac.in					
Unit Number	**Sr. No.**	**Questions / Task Description**	**Name of CO**	**Matching Program Outcome (PO)**	**Bloom's Taxonomy (BTL)**

Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Practical List-Mini Project

Subject : P21A3PYT : PYTHON PROGRAMMING

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: Dr. Ajay Patel | Email ID: ajaykumar.patel@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions / Task Description	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)
1	1	Write a Python program using NumPy to solve a system of dynamic linear equations (AX = B) using matrix inversion and matrix decomposition techniques.	CO1	1,2,3,4,5,6,7,8,11	BTL-3
1	2	Write a Python script using NumPy to calculate standard deviation, Z-score normalization, and principal diagonal transformation on a 5x5 matrix without built-in stat functions.	CO1	1,2,3,4,5,6,7,8,11	BTL-3
1	3	Write a Pandas program to clean a messy dataset: handle missing data via dynamic conditional imputation, detect and remove outliers using IQR, and encode categorical features.	CO1	1,2,3,4,5,6,7,8,11	BTL-3
1	4	Given a transactional dataset, perform multi-level grouping (groupby), calculate rolling moving averages on time-series columns, and reshape data using dynamic pivot_table.	CO1	1,2,3,4,5,6,7,8,11	BTL-4
1	**5 Case Study 1**	Financial Portfolio Performance Analyzer: Build an engine using NumPy & Pandas that ingests multi-year stock price data, calculates daily logarithmic returns, Sharpe ratio, and covariance matrices.	CO1	1,2,3,4,5,6,7,8,11	BTL-6
2	6	Design a custom Python modular package analytics_engine containing modules for advanced statistics and array transforms, featuring proper __init__.py scoping.	CO2	1,2,3,4,5,6,7,8,11	BTL-6
2	7	Develop a Banking Domain System using dynamic Object-Oriented Principles: Abstract Base Classes, Encapsulated Private Attributes, Overridden Methods, and polymorphic payment processing.	CO2	1,2,3,4,5,6,7,8,11	BTL-6
2	8	Implement a robust Exception-handling engine for a REST client simulation that handles connectivity errors, authorization failure, and rate limits using custom Exception subclasses.	CO2	1,2,3,4,5,6,7,8,11	BTL-3
2	9	Write a Python program implementing a custom generic Data Structure (e.g., Doubly Linked List or Priority Queue) with magic methods (__len__, __getitem__, __iter__).	CO2	1,2,3,4,5,6,7,8,11	BTL-3
2	**10 Mini Project 1**	E-Commerce Inventory & Order Processing System: Build an OOP-based system with classes (Product, Customer, Order, InventoryManager) supporting order queue processing and transactional rollbacks.	CO2	1,2,3,4,5,6,7,8,11	BTL-6
3	11	Write a Python program using re module to parse server access logs, extract IP addresses, timestamps, HTTP status codes, and user agents into a clean TSV file.	CO3	1,2,3,4,5,6,7,8,11	BTL-3
3	12	Write a file manipulation utility to scan dynamic multi-nested directory structures using os.walk or pathlib, find files matching specific extensions, and compile metadata summary reports.	CO3	1,2,3,4,5,6,7,8,11	BTL-3
3	13	Create a program that reads unstructured text, removes stop words, performs word frequency analysis, and exports formatted results to both CSV and JSON formats.	CO3	1,2,3,4,5,6,7,8,11	BTL-3
3	14	Build a dynamic file encryption/decryption module that reads binary files, transforms byte patterns based on a user-defined dynamic key, and safely reconstructs target files.	CO3	1,2,3,4,5,6,7,8,11	BTL-6
3	**15 Case Study 2**	Automated File Sanitizer & Archival Manager: Create an advanced system that monitors target folders, cleans sensitive metadata/PII from CSV/Text files, and compresses clean files into zip archives.	CO3	1,2,3,4,5,6,7,8,11	BTL-6
4	16	Build a custom tkinter form validation widget that validates user input (Email, Phone, Age) in real-time with dynamic visual error indicators.	CO4	1,2,3,4,5,6,7,8,11	BTL-3
4	17	Develop a dynamic Calculator GUI Application using grid() geometry manager, handling mathematical evaluation, entry history, and syntax exception alerts.	CO4	1,2,3,4,5,6,7,8,11	BTL-6
4	18	Create a dynamic Text Editor GUI using tkinter featuring file open/save dialogs, scrollable canvas/text widgets, dynamic word counter, and font styling.	CO4	1,2,3,4,5,6,7,8,11	BTL-6
4	19	Build a dynamic Interactive Data Viewer GUI featuring nested frames, combobox selection, dataset filtering inputs, and clean tabular displays (ttk.Treeview).	CO4	1,2,3,4,5,6,7,8,11	BTL-3
4	**20 Mini Project 2**	Hospital Management System Desktop App: Develop a full multi-window tkinter GUI application with dynamic patient registration forms, searchable records table, and appointment scheduler.	CO4	1,2,3,4,5,6,7,8,11	BTL-6
5	21	Implement a Linear Regression model using scikit-learn to predict housing prices. Perform feature scaling, model fitting, slope/intercept inspection, and MSE metric evaluation.	CO5	1,2,3,4,5,6,7,8,11	BTL-3
5	22	Build a Logistic Regression model on the Iris/Diabetes dataset to solve a multi-class classification problem. Generate confusion matrix and dynamic classification reports.	CO5	1,2,3,4,5,6,7,8,11	BTL-3
5	23	Implement an Unsupervised K-Means Clustering algorithm to segment customer datasets. Determine the optimal number of clusters using the Elbow Method.	CO5	1,2,3,4,5,6,7,8,11	BTL-3
5	24	Write a Python script to compare performance between two Supervised Learning classifiers (e.g., Logistic Regression vs. Decision Tree) using k-fold cross-validation.	CO5	1,2,3,4,5,6,7,8,11	BTL-4
5	**25 Mini Project 3**	End-to-End Smart Predictor Desktop Suite: Develop an integrated tkinter GUI app that ingests dynamic CSV datasets, lets users select features, trains Machine Learning models, and displays metrics.	CO5	1,2,3,4,5,6,7,8,11	BTL-6

Department of Computer Science (FCA) - Ganpat University

M.Sc.(CA & IT) - Semester I : Assignment

Subject : P21BS5CDP1 : CAREER DEVELOPMENT AND PLANNING-I

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Subject Teacher: Prof. Vidhi H. Sheth | Email ID: vidhi.sheth@ganpatuniversity.ac.in

Unit Number	Sr. No.	Questions	Name of CO	Matching Program Outcome (PO)	Bloom's Taxonomy (BTL)	Practical List	Mini Project / Case Studies

1	1	Predict the exact output of the following C code snippet and explain the step-by-step execution sequence. #include <stdio.h>
		int main() {
		int a = 5, b = 10;
		int c = a++ * ++b;
		int d = c << 2;
		printf("a = %d, b = %d, c = %d, d = %d\n", a, b, c, d);
		return 0;
		}
			CO1				

	2	What will be printed by the following program? Trace how short-circuit evaluation affects variable updates.
		#include <stdio.h>
		int main() {
		int x = 0, y = 1, z = 2;
		int res = (x++ && ++y) || ++z;
		printf("x = %d, y = %d, z = %d, res = %d\n", x, y, z, res);
		return 0;
		}
			CO1				

	3	What does the expression `(n & (n - 1))` evaluate to for `n = 16` and `n = 12`? Write a macro using this expression.
		#include <stdio.h>
		#include <stdbool.h>
		bool isPowerOfTwo(int n) {
		return (n > 0) && ((n & (n - 1)) == 0);
		}
			CO1				

	4	Determine the output of the code snippet below based on C operator precedence rules.
		#include <stdio.h>
		int main() {
		int a = 10, b = 3;
		int val = a % b + 5 > 6 ? a * b : a / b;
		printf("Result = %d\n", val);
		return 0;
		}
			CO1				

	5	Compare bitwise arithmetic (e.g. shift operators `<<`, `>>`, bitwise `&`, `|`, `^`) with standard arithmetic operators (`*`, `/`, `%`) in terms of hardware cycle performance, compiler optimization, and type safety constraints.	CO1				

	6	What is the output of the following switch case structure for `ch = 'B'`? Identify missing structural flow elements.
		#include <stdio.h>
		int main() {
		char ch = 'B';
		int count = 0;
		
		switch (ch) {
		case 'A':
		count += 10;
		case 'B':
		count += 20;
		case 'C':
		count += 30;
		break;
		case 'D':
		count += 40;
		break;
		default:
		count += 50;
		}
		printf("Count = %d\n", count);
		return 0;
		}
			CO1				

	7	Will the following loop terminate or result in an infinite loop? Explain why floating-point variables behave this way in loop control conditions.
		#include <stdio.h>
		int main() {
		float f;
		int count = 0;
		for (f = 0.0f; f != 1.0f; f += 0.1f) {
		count++;
		if (count > 20) {
		printf("Infinite loop detected! count = %d\n", count);
		break;
		}
		}
		return 0;
		}
			CO1				

	8	Trace the execution of the nested loop and give the final printed output for `sum`.
		#include <stdio.h>
		int main() {
		int sum = 0;
		for (int i = 1; i <= 3; i++) {
		for (int j = 1; j <= 3; j++) {
		if (i == j) continue;
		if (i + j > 4) break;
		sum += (i * j);
		}
		}
		printf("Sum = %d\n", sum);
		return 0;
		}
			CO1				

	9	What is the 'Dangling Else' problem in C/C++? How does compiler resolution work when `else` matches nested `if` statements without explicit brace blocks (`{}`)?	CO1				
	10	How do modern CPU branch predictors interact with `if-else` conditionals inside high-frequency loops? Explain the concept and trade-offs of Loop Unrolling (`#pragma unroll`).	CO1				

	11	Analyze the following two-pointer array iteration function. What are its time and space complexities?
		#include <stdio.h>
		void reverseArray(int arr[], int size) {
		int start = 0, end = size - 1;
		while (start < end) {
		int temp = arr[start];
		arr[start] = arr[end];
		arr[end] = temp;
		start++;
		end--;
		}
		}
			CO1				

	12	Compare the performance of `rowMajor()` versus `colMajor()` in the snippet below. Explain why one executes significantly faster.
		#include <stdio.header>
		#define ROWS 1000
		#define COLS 1000
		int arr[ROWS][COLS];
		void rowMajor() {
		long long sum = 0;
		for (int i = 0; i < ROWS; i++)
		for (int j = 0; j < COLS; j++)
		sum += arr[i][j];
		}
		void colMajor() {
		long long sum = 0;
		for (int j = 0; j < COLS; j++)
		for (int i = 0; i < ROWS; i++)
		sum += arr[i][j];
		}
			CO1				

	13	What output will be produced when traversing 3*3 matrix using boundary variables in the given spiral algorithm?
		#include <stdio.h>
		void printSpiral(int r, int c, int matrix[3][3]) {
		int top = 0, bottom = r - 1, left = 0, right = c - 1;
		while (top <= bottom && left <= right) {
		for (int i = left; i <= right; i++) printf("%d ", matrix[top][i]);
		top++;
		for (int i = top; i <= bottom; i++) printf("%d ", matrix[i][right]);
		right--;
		if (top <= bottom) {
		for (int i = right; i >= left; i--) printf("%d ", matrix[bottom][i]);
		bottom--;
		}
		if (left <= right) {
		for (int i = bottom; i >= top; i--) printf("%d ", matrix[i][left]);
		left++;
		}
		}
		}
			CO1				

	14	What happens at the hardware/OS level when array iteration exceeds allocated bounds (`arr[i]` where `i >= size`) in C? Differentiate between Stack Buffer Overflow and Heap Corruption.	CO1				

	15	What will be printed after `swapVal(x, y)` and `swapRef(&x, &y)`? Explain why `swapVal` fails to update the arguments.
		#include <stdio.h>
		void swapVal(int a, int b) {
		int temp = a;
		a = b;
		b = temp;
		}
		void swapRef(int *a, int *b) {
		int temp = *a;
		*a = *b;
		*b = temp;
		}
		int main() {
		int x = 10, y = 20;
		swapVal(x, y);
		printf("x = %d, y = %d\n", x, y);
		swapRef(&x, &y);
		printf("x = %d, y = %d\n", x, y);
		return 0;
		}
			CO1				

	16	What is the output of the program below? Why does `allocateMemory(ptr)` fail to allocate dynamic memory for `ptr` in `main()`?
		#include <stdio.h>
		#include <stdlib.h>
		void allocateMemory(int *p) {
		p = (int *)malloc(sizeof(int));
		*p = 100;
		}
		int main() {
		int *ptr = NULL;
		allocateMemory(ptr);
		if (ptr == NULL) {
		printf("ptr is still NULL!\n");
		} else {
		printf("Value: %d\n", *ptr);
		}
		return 0;
		}
			CO1				

	17	Describe how the CPU Call Stack manages parameters during Call by Value (pushing copies onto stack) versus Call by Reference (pushing memory addresses). Discuss performance implications for large structs	CO1				

	18	Predict the output of the program when `counter()` is called thrice in a loop.
		#include <stdio.h>
		void counter() {
		static int count = 10;
		int auto_var = 10;
		count ++ 5;
		auto_var ++ 5;
		printf("static = %d, auto = %d\n", count, auto_var);
		}
		int main() {
		for (int i = 0; i < 3; i++) {
		counter();
		}
		return 0;
		}
			CO1				

	19	What is the return value of `func(5)`? Trace the state of static variable `x` across recursive stack calls.
		#include <stdio.h>
		int func(int n) {
		static int x = 0;
		if (n <= 0) return 0;
		x++;
		return func(n - 1) + x;
		}
		int main() {
		printf("Result = %d\n", func(5));
		return 0;
		}
			CO1				

	20	Detail the differences between `auto`, `register`, `static`, and `extern` storage classes in C based on: (1) Storage Location, (2) Default/Initial Value, (3) Scope, and (4) Lifetime	CO1				

	21	What compiler error or warning occurs when executing the following code snippet in C? Explain the architectural reason.
		#include <stdio.h>
		int main() {
		register int x = 100;
		int *ptr = &x;
		printf("Value = %d\n", *ptr);
		return 0;
		}
			CO1				

	22	What is the output of the code snippet? Explain the difference between variable declaration (`extern int x;`) and variable definition (`int x = 50;`).
		#include <stdio.h>
		extern int x;
		int x = 50;
		int main() {
		extern int x;
		printf("x = %d\n", x);
		return 0;
		}
			CO1				

	23	Map each C storage class (`auto`, `static` initialized, `static` uninitialized, `extern`, dynamically allocated memory) to its respective memory segment in the C runtime layout (Text, Initialized Data, BSS, Heap, Stack).	CO1				

	24	Predict the output of each print statement in the code below. Explain how pointer arithmetic differs between `arr` and `&arr`.
		#include <stdio.h>
		int main() {
		int arr[] = {10, 20, 30, 40, 50};
		int *ptr1 = arr;
		int *ptr2 = (int*)(&arr + 1);
		printf("*ptr1 = %d\n", *ptr1);
		printf("*(ptr1 + 2) = %d\n", *(ptr1 + 2));
		printf("*(ptr2 - 1) = %d\n", *(ptr2 - 1));
		return 0;
		}
			CO1				

	25	Identify which statements in the code below will result in compilation errors. Explain the rule for each const qualifier placement.
		#include <stdio.h>
		int main() {
		int val1 = 10, val2 = 20;
		const int *p1 = &val1;
		int * const p2 = &val1;
		const int * const p3 = &val1;
		*p1 = 15;
		p1 = &val2;
		*p2 = 15;
		p2 = &val2;
		*p3 = 15;
		p3 = &val2;
		return 0;
		}
			CO1				

	26	#include <stdio.h>
		#include <stdlib.h>
		int main() {
		int rows = 3, cols = 3;
		int **matrix = (int **)malloc(rows * sizeof(int *));
		
		for (int i = 0; i < rows; i++) {
		matrix[i] = (int *)malloc(cols * sizeof(int));
		}
		// Freeing memory safely
		for (int i = 0; i < rows; i++) {
		free(matrix[i]);
		}
		free(matrix);
		return 0;
		}
			CO1				

	27	What is the expected output when executing `executeOperation` with function pointers?
		#include <stdio.h>
		int add(int a, int b) { return a + b; }
		int multiply(int a, int b) { return a * b; }
		void executeOperation(int (*op)(int, int), int x, int y) {
		printf("Result: %d\n", op(x, y));
		}
		int main() {
		int (*funcPtr)(int, int);
		
		funcPtr = add;
		executeOperation(funcPtr, 10, 5);
		funcPtr = multiply;
		executeOperation(funcPtr, 10, 5);
		return 0;
		}
			CO1				

	28	Explain the differences between a Dangling Pointer, Wild Pointer, Null Pointer, and Memory Leak	CO1				
	29	What is a void* pointer in C/C++? What operations are you not allowed to do with it (like pointer math or reading its value directly), and what risks come with converting (typecasting) it to other pointer types?	CO1				
	30	Write a Program for the given output . 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5	CO1				
	31	1 2 3 4 5 2 3 4 5 3 4 5 4 5 5	CO1				
	32	Chocolate Factory Problem (Array / Two-Pointer) Problem: A chocolate factory produces chocolates packed in boxes (represented by an array of integers). Some boxes are empty (0). Move all empty boxes (0s) to the end of the array while keeping the non-zero elements in their original order. Input: [4, 5, 0, 1, 9, 0, 5, 0] Output: [4, 5, 1, 9, 5, 0, 0, 0]	CO1				
	33	Write a program to swap two variables, A = 2 and B = 8, without creating a temporary variable	CO1				
	34	Given an array, count how many elements are strictly greater than all elements before them. (The 1st element is always counted)	CO1				
	35	Check whether two given strings are anagrams of each other (i.e., contain the same characters with the same frequency)	CO1				
	36	Find Kth Largest and smallest element from array in a single loop traversal O(N) without using pre-built sorting functions.	CO1				